

# A pareto-based ensemble of feature selection algorithms

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## ABSTRACT

In this paper, ensemble feature selection is modeled as a bi-objective optimization problem regarding features' relevancy and redundancy degree. The proposed method, which is called PEFS, first uses the modeled bi-objective optimization problem to find the non-dominated features based on the decision matrix constructed by different feature selection algorithms. In the second step, the found non-dominated features are sorted using the crowding distance in the bi-objective space. These sorted features remove from the feature space, and the process of finding the non-dominated features will continue until all the features are sorted. To illustrate the optimality and efficiency of the proposed method, we have compared our approach with some ensemble feature selection methods and basic algorithms used in the ensemble process. The results show that our method in terms of accuracy and F-score is superior to other similar methods and performs in a short running-time.

## 1. Introduction

In many real-world applications, the data include irrelevant and redundant features. The irrelevant features do not affect achieving the output, and the redundant features are features that are almost a mix of other features and do not provide new information. The existence of such features in datasets makes learning algorithms time-consuming, and it also affects their performance. Feature selection (Dowlatshahi & Derhami, 2017; Dowlatshahi, Derhami, & Nezamabadi-pour, 2018) is used as an effective way to deal with these problems. Feature selection as a dimensionality reduction technique selects essential features and removes the irrelevant and redundant ones from the dataset. This procedure can reduce the dimensionality of data, computational cost, time complexity, storage complexity, and increase classification accuracy (Cai, Luo, Wang, & Yang, 2018; Dowlatshahi & Rezaeian, 2016; Miao & Niu, 2016; Mlambo, 2016). From two perspectives, we can categorize feature selection methods: supervision and search strategy (Li et al., 2018). From the supervision perspective, these methods can be divided into three main classes: supervised, unsupervised, and semi-supervised (Zhang, Nie, Li, & Wei, 2019). A supervised feature selection method selects features based on label information. Supervised methods are applied to Single-Label (SL) and Multi-Label (ML) data (Bayati,

Dowlatshahi, & Paniri, 2020; Hashemi & Dowlatshahi, 2020; Hashemi, Dowlatshahi, & Nezamabadi-pour, 2020a,b; Hashemi, Dowlatshahi, & Nezamabadi-Pour, 2021; Paniri, Dowlatshahi, & Nezamabadi-pour, 2020). Unlike SL, each sample contains more than one label in ML data, and the correlation between features and labels are considered feature relevancy (Pereira, Plastino, Zadrozny, & Merschmann, 2018; Zhang et al., 2019). Unsupervised methods refer to the methods that evaluate the features based on training samples without class labels (Zhang et al., 2019). Semi-supervised methods work with a type of data that includes some labeled samples (Sheikhpour, Sarram, Gharaghani, & Chahooki, 2017).

Based on the search strategy, feature selection methods also can be partitioned into three main groups: filter, wrapper, and embedded. In filter methods, the statistical measures are used to evaluate the features Independent of the learning process (Dowlatshahi, Derhami, & Nezamabadi-pour, 2017; Venkatesh & Anuradha, 2019; Zhou, Zhang, Zhou, Guo, & Ma, 2021). In (Hancer, Xue, & Zhang, 2018), a new criterion is proposed, which tries to rank the features using ReliefF (Reyes, Morell, & Ventura, 2015) and Fisher-score (Hancer et al., 2018) and also provided the mutual relevance between features and the class labels. On the other hand, wrapper methods evaluate the features in the learning procedure (Nouri-Moghaddam, Ghazanfari, & Fathian, 2021). These

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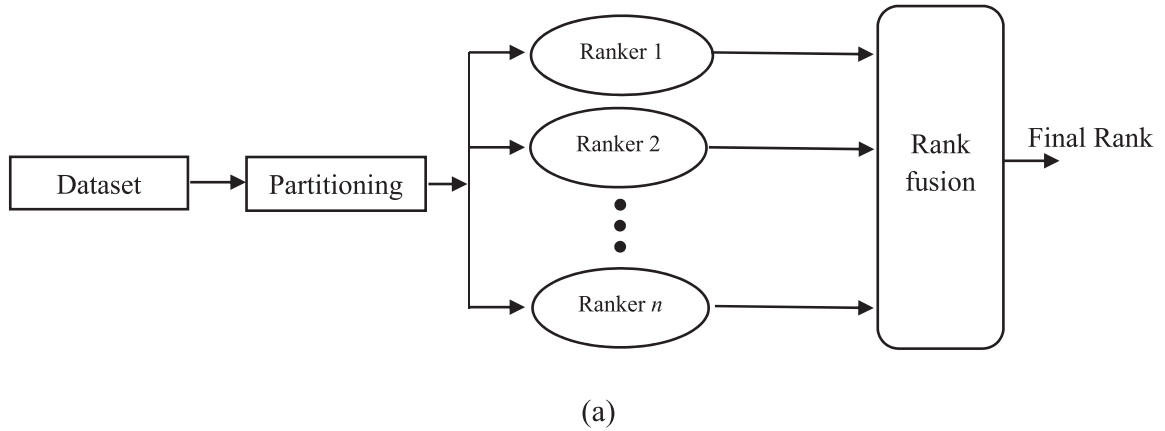
methods assess the feature subsets to determine the best values according to their predictive power (Miao & Niu, 2016). In (Jadhav, He, & Jenkins, 2018), a wrapper feature selection method called IGDFS is presented for financial credit scoring. IGDFS used information gain to rank the features and then identified the top  $m$  features using the GA wrapper algorithm. Finally, embedded methods seek to address the weaknesses of the two previous techniques and use the strengths of both (Li et al., 2018). Zhu and Yang (2018) proposed an unsupervised feature selection algorithm that uses an embedded strategy to set up a ranking system by maximizing the distances between instances from different clusters.

Alongside these methods, a new method called ensemble feature selection has recently been introduced. In this method, the output of several feature selection methods is aggregated to obtain the final feature subset. The rationale for this approach is based on the old proverb “Two heads are better than one” and on the fact that the variety of different methods makes the feature selection method work better than only one method. Each feature selection algorithm may find feature subsets that can be considered local optima in the feature subsets space. Ensemble feature selection can be useful to address this problem (Bolón-Canedo & Alonso-Betanzos, 2019).

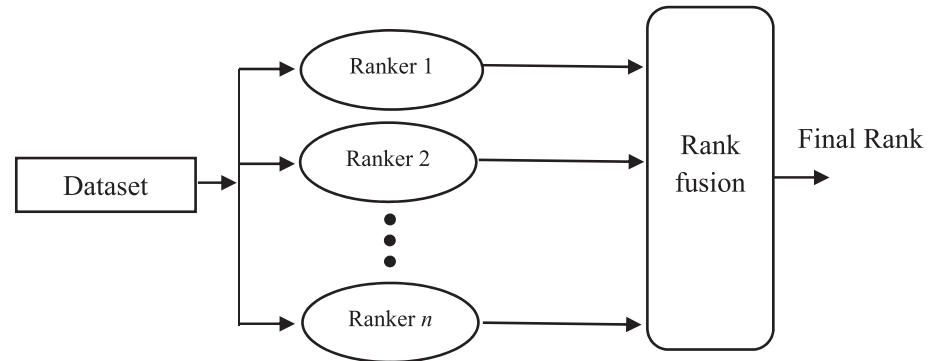
An optimization process is finding the best solution out of all the possible solutions. Optimization is widely used in real-world issues (Dowlatshahi & Derhami, 2017; Dowlatshahi, Derhami, & Nezamabadi-Pour, 2020; Dowlatshahi, Nezamabadi-Pour, & Mashinchi, 2014; Dowlatshahi & Nezamabadi-Pour, 2014; Momeni, Dowlatshahi, Omidinasab, Maizir, & Armaghani, 2020; Momeni, Yarivand, Dowlatshahi, & Armaghani, 2021; Marjan Kuchaki Rafsanjani & Dowlatshahi, 2012; M. K. Rafsanjani, Dowlatshahi, & Nezamabadi-Pour, 2015). Many optimization problems consist of multiple objectives that need to be optimized

simultaneously. These problems are called multi-objective optimization problems and occur in many engineering and medical tasks (Li, Yang, & Liu, 2015). The conflict between objectives leads to a set of trade-off solutions called the Pareto optimal set instead of an optimal one. The solutions in the Pareto optimal set are called non-dominated solutions. A non-dominated solution is a solution that is not worse than any other solution and is better than at least one solution, which means that there is no solution to dominate it. One of the most important methods to solve multi-objective optimization problems is a Multi-objective Evolutionary Algorithm (MOEA). In general, MOEAs are known as iterative algorithms that obtain the Pareto optimal set using the concept of Pareto dominance (Von Lücken, Barán, & Brizuela, 2014). In (Yuan, Xu, Wang, & Yao, 2016), a many-objective evolutionary method is presented. This method selects the solutions based on a one-by-one strategy that uses a convergence indicator. In many-objective problems that refer to a multi-objective problem with more than three objectives, if the number of objectives increases, Pareto dominance loses its effectiveness. Based on experiments, the Pareto-based methods have shown excellent performance with two or three objectives. The main reason is that with the increase in the number of the objective, almost every solution becomes non-dominated that loses the performance of Pareto-based algorithms (Von Lücken et al., 2014; Yuan et al., 2016).

This article models the ensemble feature selection problem to a Pareto-based optimization problem with two objectives. In this method, we use the results of two ensemble algorithms and the concept of Pareto dominance to give a higher ranking to the features that are not dominated by other features. We first combined the results obtained by four different feature selection methods using two aggregation methods in the proposed method. Finally, we evaluated these new results using a bi-objective optimization. The non-dominant features are ranked in the bi-



(a)



(b)

Fig. 1. Block diagram of (a) Homogeneous and (b) Heterogeneous ensemble FS.

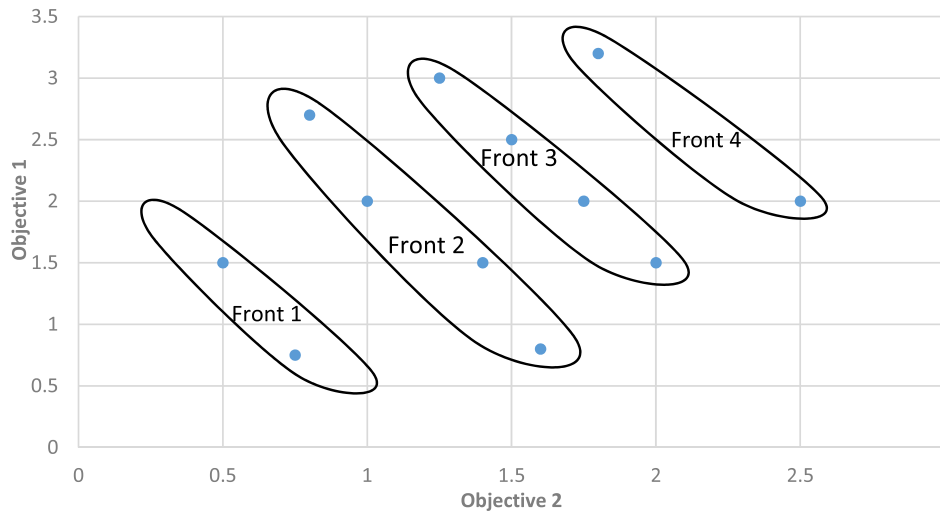


Fig. 2. Example of sorting 13 solutions into 4 fronts in a bi-objective minimization problem.

objective space according to their crowding distance. These features are removed from the space so that other features are ranked based on this strategy. The primary feature selection methods used in this algorithm are a combination of relevancy and redundancy methods. The proposed method can balance these two properties and give a higher ranking to the features that have the most relevancy with the class label and the least redundancy. To test the performance of the proposed method, seven real-world data sets, including five biological data and two image data with different dimensions, have been used. To illustrate the optimality and efficiency of the proposed method, we have compared our approach with some similar techniques and base FS algorithms on different real-world datasets. The results indicate that our proposed method yields better results than the other methods in terms of accuracy, F-score, and run-time.

An overview of our proposed method is as follows:

- The user can determine the number of selected features,
- This algorithm is an ensemble of filter methods, and
- A non-dominated sorting strategy is used as the first measure and the crowding distance is used as the secondary measure.

The structure of this paper is organized as follows: Section 2 deals with reviewing related methods. In Section 3, the fundamental concepts used in the proposed method, and Section 4 represents the proposed method in detail. Section 5 includes the experimental results, and Section 6 represents the conclusion and expression of future works.

## 2. Related works

In (Seijo-Pardo, Porto-Díaz, Bolón-Canedo, & Alonso-Betanzos, 2017), two different homogeneous and heterogeneous approaches are presented. They used several aggregators for combining the results of various rankers. In (Ben Brahim & Limam, 2018), an ensemble feature selection method is presented for high dimensional data. In this method, the combination of different rankers is based on the reliability assessment-based aggregation (RAA) technique. Das, Das, and Ghosh (2017) proposed a new ensemble FS method based on a bi-objective genetic algorithm. This algorithm tries to find the best subset of features using dynamic mating and considers rough set theory and information theory as their two objective functions. Ansari, Ahmad, and Doja (2019), proposed an ensemble FS method for sentiment classification. They used the concept of hesitant fuzzy sets to combine the results of different filter FS methods. This approach selects top-k ranked features based on the relevancy score. In Ebrahimipour and Eftekhari (2017)s, a technique based on Maximum Relevancy and Minimum Redundancy

(MRMR) is presented for ensemble FS using Hesitant Fuzzy Sets (HFSs). This algorithm is a filter-based method, and the results of different rankers are combined using the concept of information energy of HFSs.

In Drotár, Gazda, and Vokorokos (2019), several ensemble FS methods are presented based on some basic techniques like max, min, ... and election methods. They also performed several clustering-based methods using the mean-shift algorithm. Ensemble-based Filter Feature Selection (EFFS) (Tian, Zhang, Wang, Geng, & Wang, 2020) is a heterogeneous approach for wearable sensor-based human activity recognition. In this method, the aggregation of four filter feature ranking methods includes information gain, gain ratio, chi-squared, and ReliefF, are considered to achieve the final feature set. The aggregation method used in EFFS is a weighted mean, and the weight for each method is obtained from multiple experiments. EFS-MI (Hoque, Singh, & Bhattacharyya, 2018) is an ensemble feature selection algorithm that used the aggregation ranks assigned to features by filter feature ranking methods. This method used gain ratio, information gain, chi-squared, ReliefF, and symmetric uncertainty as the based feature ranking methods. The ensemble process of this method is based on feature-class and feature-feature mutual information. Recently other ensemble feature selection procedures are proposed for high dimensional datasets with low number of instances. In this article, some serial and parallel techniques have been offered (Tsai & Sung, 2020). Another approach is presented to deal with the sensitivity of ensemble feature selection algorithms and minimizing the training error. The NSGA-III is used to find the optimal feature subsets (Ng, Tuo, Zhang, & Kwong, 2020).

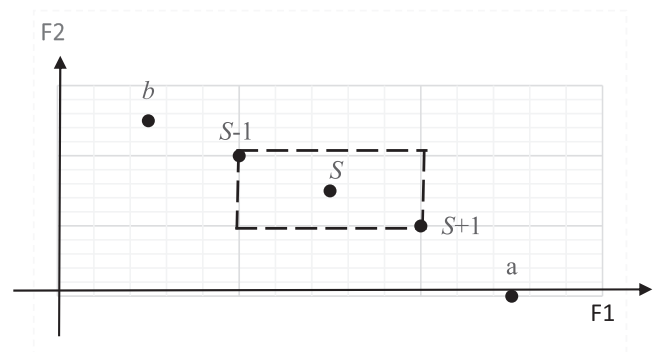


Fig. 3. Crowding distance of point S in a bi-objective space.

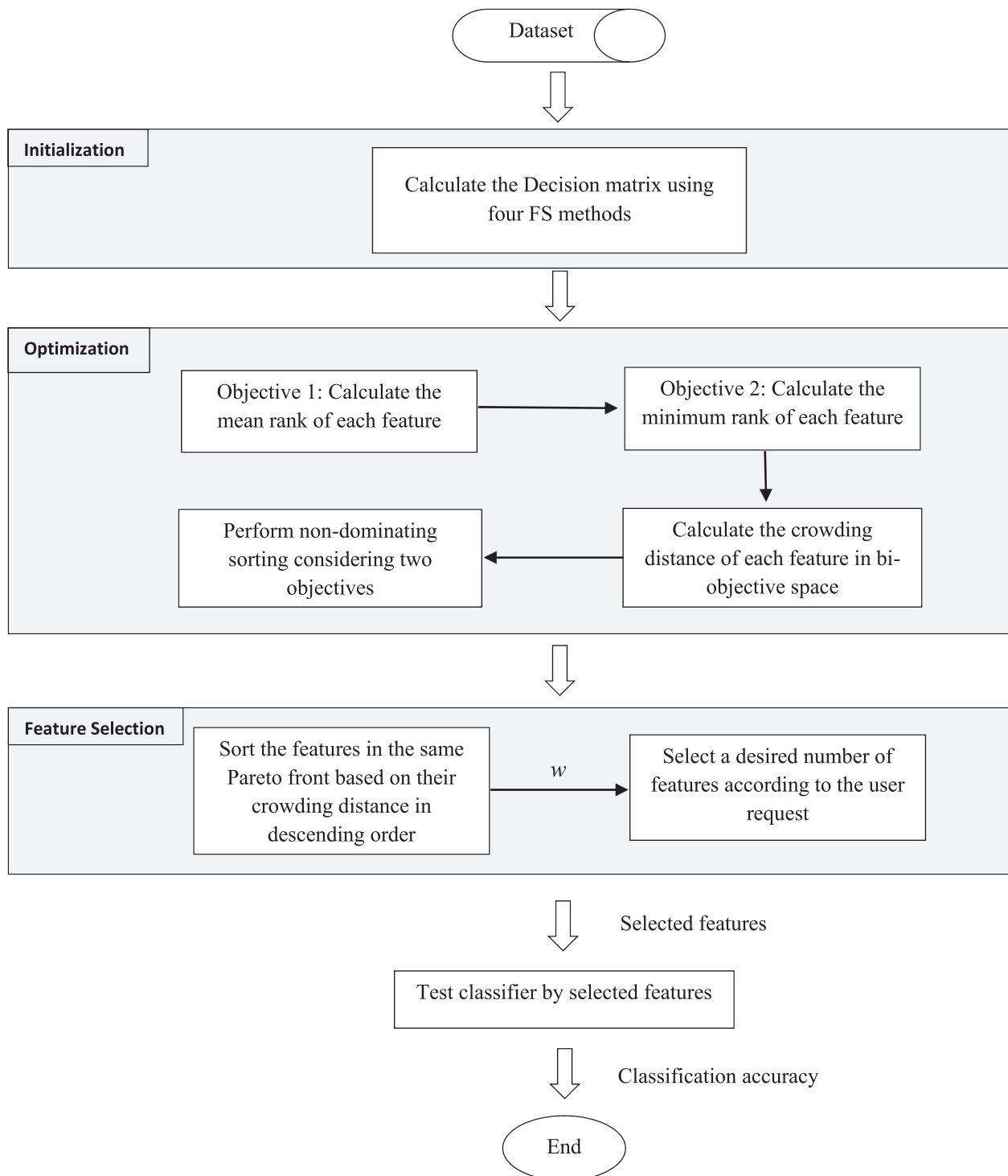


Fig. 4. Graphical abstract of the proposed method.

### 3. Fundamental concepts

#### 3.1. Ensemble feature selection

Ensemble feature selection is the process of combining multiple models instead of one model. The rationale for this approach is based on the old proverb “Two heads are better than one” and on the fact that the variety of different methods makes the feature selection method work better than only one method. Various studies have shown that these methods usually achieve better accuracy than individual methods because of the diversity of these approaches and control of variance.

Ensembles have recently been used in machine learning techniques such as supervised classification, regression, and optimization. In the area of feature selection, these methods have also recently been widely used.

Ensemble FS methods can be categorized into two main groups: The first group is called homogeneous ensembles, which exploit data diversity. In these methods, the data is partitioned into multiple partitions, and a feature selection method is implemented on each partition. Finally, all partitions’ results are combined to achieve the final feature subset (Fig. 1(a)). The second group is called heterogeneous ensembles, which exploit the diversity of functions. In these methods, multiple feature selection methods are executed on the data, and the results of

$$X = \begin{bmatrix} 1 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 1 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 \\ 1 & 1 & 0 & 0 & 0 & 0 & 0 & 0 & 0 & 1 & 1 & 0 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 1 \\ 0 & 0 & 0 & 1 & 1 & 0 & 0 & 1 & 0 & 0 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 & 0 & 1 \end{bmatrix} \quad Y = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

**a) Sample dataset**

$$R1 = \{3,7,13,19,1,2,4,5,6,8,9,10,11,12,14,15,16,17,18,20\}$$

$$R2 = \{2,3,4,5,7,10,13,14,17,19,6,8,9,12,15,16,18,1,11,20\}$$

$$R3 = \{1,8,11,12,16,18,2,4,5,10,14,17,6,9,15,20,3,7,13,19\}$$

$$R4 = \{2,4,6,10,14,17,1,6,8,9,11,12,15,16,18,20,3,7,13,19\}$$

**b) Ranking vectors based on 4 rankers**

$$P = \begin{bmatrix} 3 & 2 & 1 & 2 \\ 7 & 3 & 8 & 4 \\ 13 & 4 & 11 & 6 \\ 19 & 5 & 12 & 10 \\ 1 & 7 & 16 & 14 \\ 2 & 10 & 18 & 17 \\ 4 & 13 & 2 & 1 \\ 5 & 14 & 4 & 6 \\ 6 & 17 & 5 & 8 \\ 8 & 19 & 10 & 9 \\ 9 & 6 & 14 & 11 \\ 10 & 8 & 17 & 12 \\ 11 & 9 & 6 & 15 \\ 12 & 12 & 9 & 16 \\ 14 & 15 & 15 & 18 \\ 15 & 16 & 20 & 20 \\ 16 & 18 & 3 & 3 \\ 17 & 1 & 7 & 7 \\ 18 & 11 & 13 & 13 \\ 20 & 20 & 19 & 19 \end{bmatrix} \quad D = \begin{bmatrix} 5 & 18 & 1 & 7 \\ 6 & 1 & 7 & 1 \\ 1 & 2 & 17 & 17 \\ 7 & 3 & 8 & 2 \\ 8 & 4 & 9 & 3 \\ 9 & 11 & 13 & 8 \\ 2 & 5 & 18 & 18 \\ 10 & 12 & 2 & 9 \\ 11 & 13 & 14 & 10 \\ 12 & 6 & 10 & 4 \\ 13 & 19 & 3 & 11 \\ 14 & 14 & 4 & 12 \\ 3 & 7 & 19 & 19 \\ 15 & 8 & 11 & 5 \\ 16 & 15 & 15 & 13 \\ 17 & 16 & 5 & 14 \\ 18 & 9 & 12 & 6 \\ 19 & 17 & 6 & 15 \\ 4 & 10 & 20 & 20 \\ 20 & 20 & 16 & 16 \end{bmatrix} \quad F_1 = \begin{bmatrix} 7.75 \\ 3.75 \\ 9.25 \\ 5 \\ 6 \\ 10.25 \\ 10.75 \\ 8.25 \\ 12 \\ 8 \\ 11.5 \\ 11 \\ 12 \\ 9.75 \\ 14.75 \\ 13 \\ 11.25 \\ 14.25 \\ 13.5 \\ 18 \end{bmatrix} \quad F_2 = \begin{bmatrix} 1 \\ 1 \\ 1 \\ 2 \\ 3 \\ 8 \\ 2 \\ 2 \\ 10 \\ 4 \\ 3 \\ 4 \\ 3 \\ 5 \\ 13 \\ 5 \\ 6 \\ 6 \\ 4 \\ 16 \end{bmatrix}$$

**c) Feature Ranking Matrix ( $P$ )**

**d) Decision Matrix ( $D$ )**

**e) First objective**

**f) Second objective**

|             |          |          |          |          |          |          |          |      |          |          |    |          |          |          |          |      |      |          |          |          |
|-------------|----------|----------|----------|----------|----------|----------|----------|------|----------|----------|----|----------|----------|----------|----------|------|------|----------|----------|----------|
| Feature num | 1        | 2        | 3        | 4        | 5        | 6        | 7        | 8    | 9        | 10       | 11 | 12       | 13       | 14       | 15       | 16   | 17   | 18       | 19       | 20       |
| ParetoNum   | 2        | 1        | 3        | 2        | 3        | 6        | 4        | 3    | 7        | 4        | 5  | 5        | 6        | 5        | 9        | 7    | 6    | 8        | 7        | 10       |
| $d$         | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | $\infty$ | 2    | $\infty$ | $\infty$ | 2  | $\infty$ | $\infty$ | $\infty$ | $\infty$ | 2    | 2    | $\infty$ | $\infty$ | $\infty$ |
| R           | 2        | 1        | 3        | 2        | 3        | 6        | 4        | 3.33 | 7        | 4        | 5  | 5.33     | 6        | 5        | 9        | 7.33 | 6.33 | 8        | 7        | 10       |
| Rank        | 2        | 1        | 4        | 3        | 5        | 12       | 7        | 6    | 15       | 8        | 9  | 11       | 13       | 10       | 19       | 17   | 14   | 18       | 16       | 20       |

**g) The final rank of each feature**

**Fig. 5.** Example of the steps of the proposed method.

**Table 1**

Description of the used datasets.

| Reference                | Features | Instances | Dataset |
|--------------------------|----------|-----------|---------|
| (Li et al., 2018)        | 2000     | 62        | Colon   |
| (Li et al., 2018)        | 4435     | 50        | Glioma  |
| (Shipp et al., 2002)     | 5470     | 77        | DLBCL   |
| (Samaria & Harter, 1994) | 1025     | 400       | ORL     |
| (Li et al., 2018)        | 3313     | 203       | Lung    |
| (Bache & Lichman, 2013)  | 257      | 1593      | Semeion |
| (Shipp et al., 2002)     | 7130     | 77        | Shipp   |

these FS methods aggregate to find the best subset of features (Fig. 2(b)) (Bolón-Canedo & Alonso-Betanzos, 2018, 2019).

### 3.2. Pareto-based solutions

In multi-objective optimization problems (MOPs), the conflict between objectives leads to a set of trade-off solutions called the Pareto optimal set instead of an optimal one. This paper used minimizing optimization for the Pareto optimization concepts without losing generality (Talbi, 2009; Von Lücken et al., 2014). A MOP is formulated as follows:

$$\begin{cases} \min F(x) = (f_1(x), f_2(x), \dots, f_n(x)), \\ s.t. x \in S \end{cases} \quad (1)$$

where  $n(n \geq 2)$  denotes the number of objectives,  $x = (x_1, \dots, x_k)$  is the decision variables vector, and  $S$  represents the set of possible solutions. Finally,  $F(x) = (f_1(x), f_2(x), \dots, f_n(x))$  is the objectives vector that should be optimized. When our objective space includes two objectives, finding optimal solutions is called a bi-objective optimization problem. Also, an optimization problem with more than three goals is called a many-objectives optimization problem. Since we consider the MOP as a minimization problem, the vector  $F$  can be defined as a benefit function that estimates each solution's quality.

In multi-objective optimization problems, the following concepts are widely discussed:

**Pareto Dominance:** An objective vector  $u = (u_1, \dots, u_n)$  is said to dominate  $v = (v_1, \dots, v_n)$  (denoted by  $u \prec v$ ) if and only if no component of  $v$  is smaller than the corresponding element of  $u$  and at least one element of  $u$  is strictly smaller, that is (Ehrgott, 2000):

$$\forall i \in \{1, \dots, n\} : u_i \leq v_i \wedge \exists i \in \{1, \dots, n\} : u_i < v_i. \quad (2)$$

Pareto dominance is a measure of MOP efficiency and uses to say when a solution is better than the other. It is critical in MOP because different solutions can have an advantage over each other on different objectives (Duleba & Moslem, 2019).

**Pareto Optimal Set:** For a solution in Eq. (2),  $x^* \in S$ , if no solution  $x' \in S$  satisfies  $x' \prec x^*$ , then  $x^*$  is an optimal solution. All the solutions that satisfy the following condition are members of the Pareto optimal set (Bozóki & Fülöp, 2018).

**Pareto Optimal Front:** The projection of the Pareto optimal set on the objective space is considered as Pareto optimal front.

### 3.3. Non-dominated sorting

As discussed in the previous section, the Pareto dominance concept is a powerful method to deal with multi-objective optimization problems. Non-dominated sorting is a very effective strategy for finding Pareto optimal solutions. In this strategy, each solution in objective space is assigned to different fronts based on their dominant relationships. If we consider the number of solutions equal to  $P$ , then this population of solutions can be categorized onto  $K$  fronts, where  $F_i, i = 1, \dots, K$  is the front number. So, all the non-dominated solutions are assigned to  $F_1$  and the non-dominated solutions in  $P - F_1$  are also assigned to  $F_2$ . This process is repeated until all the solutions are assigned by front numbers (Gustavsson & Syberfeldt, 2018; Zhang, Tian, Cheng, & Jin, 2015). Fig. 2 demonstrates an example of sorting 13 solutions into 4 fronts in a bi-objective minimization problem using non-dominated sorting.

This paper has used a non-dominated sorting algorithm called the Efficient Non-dominated Sort Sequential Strategy (ENS-SS) (Zhang et al., 2015). Unlike other non-dominated sorting strategies, the front number of all solutions is not determined as a whole in ENS-SS. But it applies the solutions to the front numbers one by one. It can avoid the duplication of comparisons and thus reduces computational complexity. In ENS-SS, at first, the first solution is assigned to front 1. Then, the following solution is compared with the first one. If they do not dominate each other, it is also assigned to front 1. Otherwise, the front 2 is created. If the second solution dominates the first one, then the first solution is moved to front 2, and the second one is assigned to front 1. In vice versa, the second solution is assigned to front 2. The pseudocode of the ENS-SS is presented in algorithm 1. If we assume that the ENS-SS algorithm is applied to  $N$  solutions with  $M$  objectives, then the

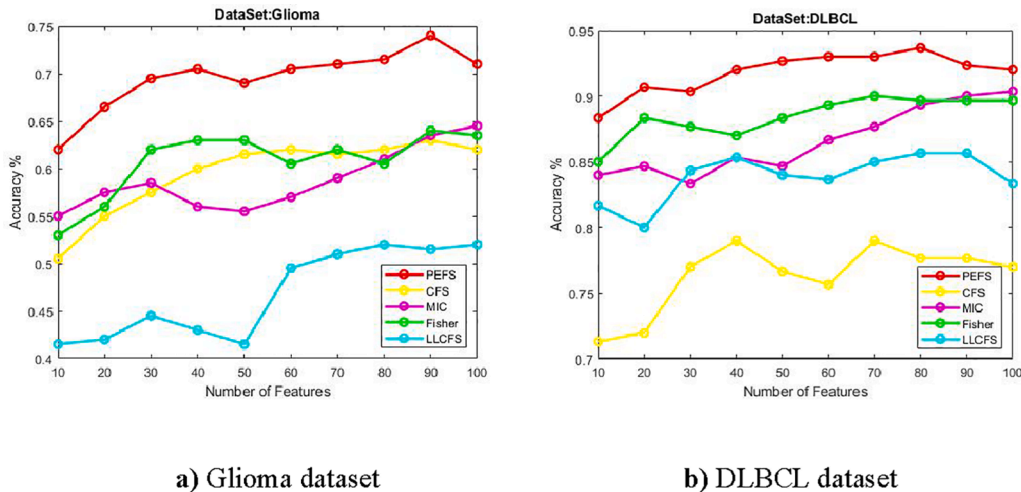
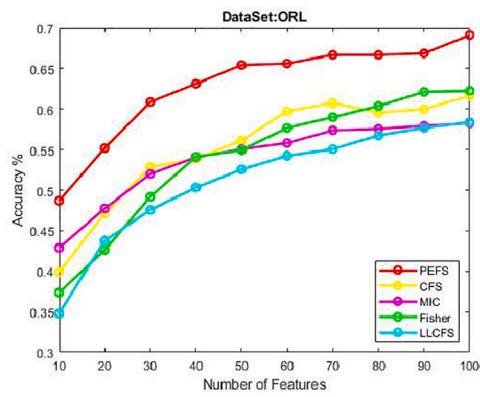
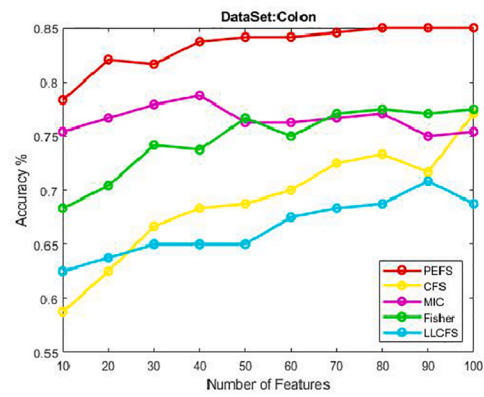


Fig. 6. Results for accuracy metric.

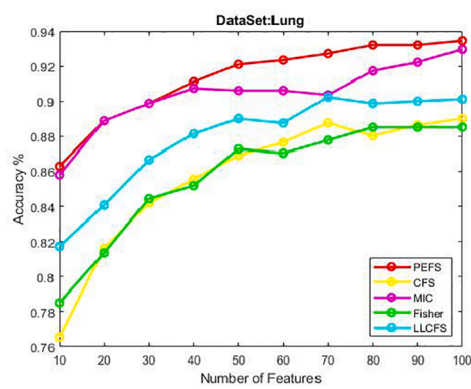




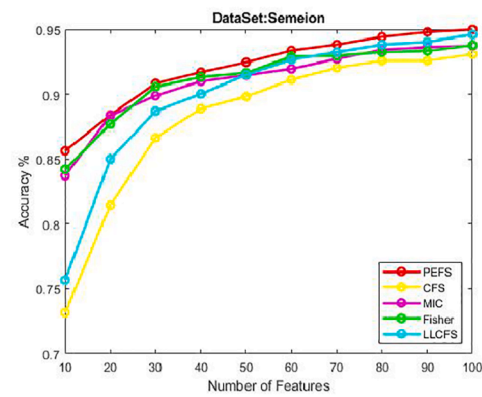
c) ORL dataset



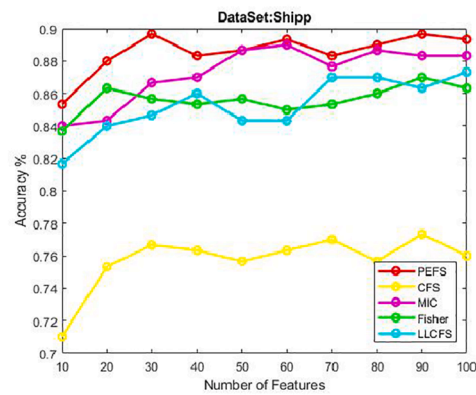
d) Colon dataset



e) Lung dataset

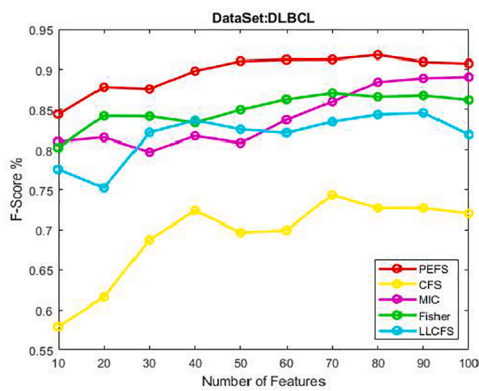


f) Semeion dataset

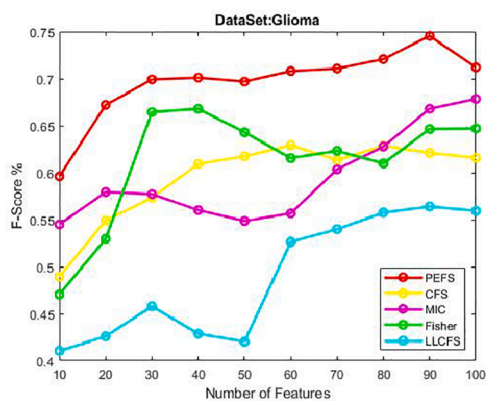


g) Shipp dataset

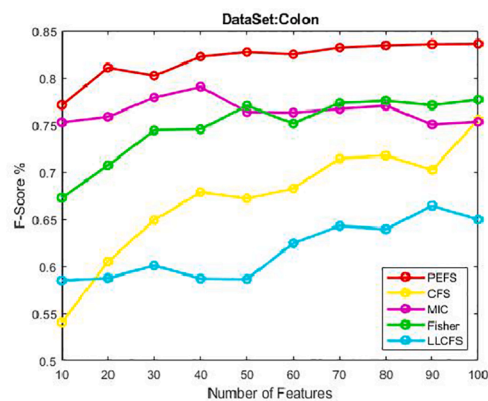
Fig. 6. (continued).



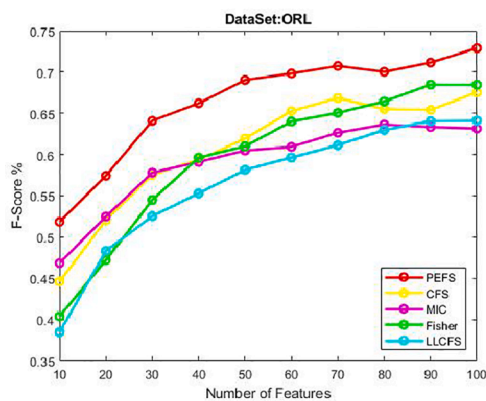
a) DLBCL dataset



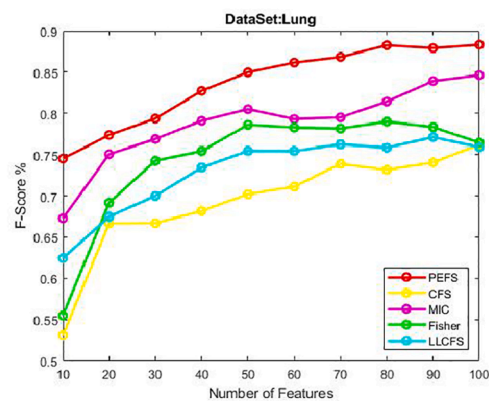
b) Glioma dataset



c) Colon dataset



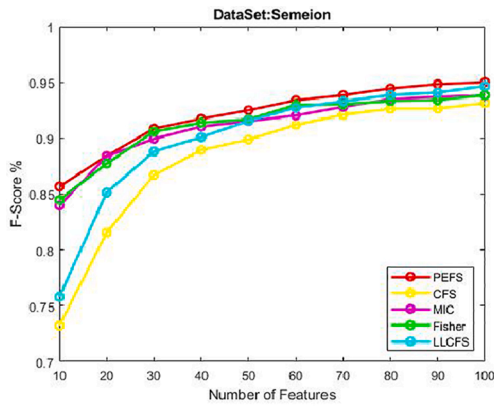
d) ORL dataset



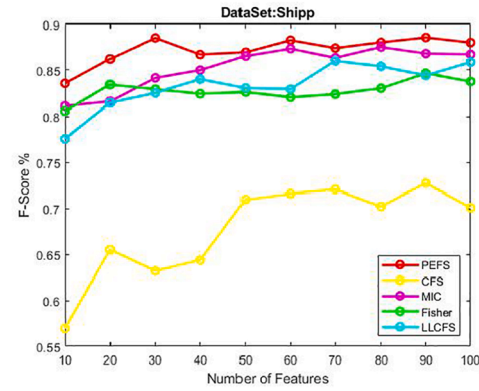
e) Lung dataset

Fig. 7. Results for F-score metric.





f) Semeion dataset



g) Shipp dataset

Fig. 7. (continued).

computational complexity is  $O(MN\sqrt{N})$  (Zhang et al., 2015).

Algorithm 1: ENS-SS

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Input: Solution  $P[n]$ , the set of fronts  $F$   
Output: the front number of solution  $P[n]$

1.  $\times = \text{size}(F)$  # the number of fronts having been found
2.  $kmin = 0$  # the lower bound for checking
3.  $kmax = \times$  # the upper bound for checking
4.  $k = \lfloor (kmax + kmin)/2 + 1/2 \rfloor$  # the front now checked
5. While true do
6. compare  $P[n]$  with the solutions in  $F[k]$  starting from the last one and ending with the first one
7. If  $F[k]$  has no solution dominating  $P[n]$  then
8. If  $k == kmin + 1$  then
9. return  $k$  # move  $P[n]$  to  $F[k]$
10. break
11. else
12.  $kmax = k$
13.  $k = \lfloor (kmax + kmin)/2 + 1/2 \rfloor$
14. end if
15. else
16.  $kmin = k$
17. if  $(kmax == kmin + 1)$  and  $(kmax < \times)$  then
18. return  $kmax$  # move  $P[n]$  to  $F[kmax]$
19. break;
20. else if  $kmin == \times$  then
21. return  $\times + 1$  # move  $P[n]$  to a new front
22. break;
23. else
24.  $k = \lfloor (kmax + kmin)/2 + 1/2 \rfloor$
25. end if
26. end if
27. end while

---

### 3.4. Crowding distance

The crowding distance of a solution represents the density of other solutions surrounding it. Fig. 3 shows the calculation of the crowding distance of solution  $S$  in a bi-objective space. This distance is based on estimating the largest cubed enclosing  $S$  so that any other solution is not included. The crowding distance of points  $a$  and  $b$  are set to infinite. Then, the crowding distance is obtained by normalizing the calculated distance with the difference between the maximum and minimum distance on the same Pareto front (Raquel & Naval, 2005; Xu, Chong, Karaguzel, & Lam, 2016).

## 4. Proposed method

In this section, we discuss our proposed algorithm in detail. This algorithm is a filter-based heterogeneous algorithm. In Fig. 4, a graphical abstract of the proposed method is presented. In this method, we

used the ensemble of four FS methods to model the ensemble FS problem to a Pareto-based optimization procedure. Algorithm 2 shows the step-by-step process of the proposed method.

### 4.1. Motivation

Feature selection algorithms usually find feature subsets that can be considered local optima in the feature subsets space. To overcome this issue, ensemble strategies are very useful. Ensemble feature selection is the process of aggregating the output of multiple FS algorithms to improve their robustness and also the learning accuracy.

The real-world problems are usually a combination of multiple objectives that need to be optimized simultaneously. In these optimization problems, a confliction between the objectives often leads to a set of trade-off solutions called the Pareto optimal set instead of an optimal one. A non-dominated solution is a solution that is not worse than any other solution and is better than at least one solution, which means that there is no solution to dominate it. Based on experiments, the performance of Pareto-based methods is excellent with two or three objectives. Due to the increase in the number of the objective, almost every solution becomes non-dominated, which causes a drop in the performance of Pareto-based algorithms. Considering these issues, we decided to use a Pareto-based approach for ensemble feature selection based on bi-objective strategy in this paper.

Algorithm 2: A Pareto-based ensemble of feature selection algorithms (PEFS)

---

Input:  $N \times M$  Feature data matrix  $X$ ,  $N \times 1$  label vector  $Y$   
Output: Feature ranking vector  $W$

1.  $w = \emptyset$ ;
2.  $R1$  = Feature ranking vector based on CFS method
3.  $R2$  = Feature ranking vector based on Fisher – Score method
4.  $R3$  = Feature ranking vector based on LLCFS method
5.  $R4$  = Feature ranking vector based on the MIC method
6.  $P = [R1, R2, R3, R4]$ ;
7.  $[m, n] = \text{size}(P)$
8.  $D = \text{zeros}(m, n)$
- # Construct the Preference-Matrix ( $P$ ) based on feature ranks
9. For  $i = 1: n$
10. For  $j = 1: m$
11.  $temp = P(j, i)$
12.  $D(temp, i) = j$
13. end
14. end
15.  $F1 = \text{mean}(D, 2)$ ; # Calculate the average value of each row (feature)
16.  $F2 = \text{min}(D, 2)$ ; # Find the minimum value for each row (feature)
17. while  $(F = 0)$
18. Perform non-dominated sorting considering  $F1$  and  $F2$  vectors as two objective functions
19.  $S$  = non-dominated feature subset determined by Algorithm 1

---

(continued on next page)

(continued)

Algorithm 2: A Pareto-based ensemble of feature selection algorithms (PEFS)

---

```

20.  ParetoNum{features ∈ S} = k
21.  k = k + 1
22.  F = F · S
23.  end while
24.  # Sort the features with the same Pareto-num according to their crowding
    distance in objective space
25.  d = Calculate the crowding distance of each solution (feature)
26.  R = ParetoNum + 1/(1 + d)
27.  w = sort the features based on their values in R in ascending order

```

---

#### 4.2. Proposed algorithm

Now we describe the steps of algorithm 2 as our proposed method. In the first step, we define an empty vector ( $w$ ) as a feature ranking vector to add the features to it.

In steps 2–5 of the algorithm, we have applied four FS methods (Fisher-Score (Duda, Hart, & Stork, 2001), MIC (Duda et al., 2001), LLCFS (Zeng & Cheung, 2011), and CFS (Michalak & Kwasnicka, 2010)) to a dataset to achieve the feature ranking vectors (R1 to R4). We tried to use a combination of redundancy-based and relevancy-based FS methods to achieve the most relevant and less redundant feature subset. To do this, we used the CFS and LLCFS methods as our redundancy-based methods and the other FS methods as our relevancy-based metrics. Each ranking vector represents the ranking of features based on the following FS method. This means that the first element of each vector is the index of the highest rank feature. These vectors are constructed as follows:

$$R1 = \{r_{1,1}, r_{1,2}, \dots, r_{1,M}\} \#CFS \quad (3)$$

$$R2 = \{r_{2,1}, r_{2,2}, \dots, r_{2,M}\} \#Fisher - score \quad (4)$$

$$R3 = \{r_{3,1}, r_{3,2}, \dots, r_{3,M}\} \#LLCFS \quad (5)$$

$$R4 = \{r_{4,1}, r_{4,2}, \dots, r_{4,M}\} \#MIC \quad (6)$$

In step 6, we constructed the  $P$  matrix as our decision matrix, in which each column represents the vectors obtained from the last steps, and the rows refer to the features. In step 7, we calculated the  $P$  matrix size, which  $m$  represents the rows (features), and  $n$  is the number of columns (FS methods). The structure of the  $P$  matrix is as follows:

$$P = \begin{bmatrix} r_{11} & r_{21} & r_{31} & r_{41} \\ r_{12} & r_{22} & r_{32} & r_{42} \\ \vdots & \vdots & \vdots & \vdots \\ r_{1M} & r_{2M} & r_{3M} & r_{4M} \end{bmatrix} \quad (7)$$

In step 8 to 14, we assigned each feature its corresponding rank according to the  $P$  matrix to construct the decision matrix ( $D$ ) as follows:

$$D = \begin{bmatrix} d_{11} & d_{21} & d_{31} & d_{41} \\ d_{12} & d_{22} & d_{32} & d_{42} \\ \vdots & \vdots & \vdots & \vdots \\ d_{1M} & d_{2M} & d_{3M} & d_{4M} \end{bmatrix} \quad (8)$$

For example, if feature 8 is the highest rank feature according to R1 (actually if  $r_{1,1} = 8$ ), then  $d_{18} = 1$ . This means that feature 8 has assigned rank 1 by FS 1.

As we discussed earlier, we used a bi-objective strategy to rank the features. Therefore, we define our objective functions considering the optimization problem as a minimization approach:

**Mean:** As step 15, we consider the mean rank value of each feature to set as our first objective. So, we calculate the mean value in each row of the  $D$  matrix:

$$F1 = [\text{mean}(D(1, :)), \text{mean}(D(2, :)), \dots, \text{mean}(D(M, :))] \quad (9)$$

**Min:** As our second objective, we consider the best rank assigned to each feature. So, we calculate the min value in each row of the  $D$  matrix:

$$F2[\min(D(1, :)), \min(D(2, :)), \dots, \min(D(M, :))] \quad (10)$$

In steps 17 to 23 of Algorithm 2, a non-dominated sorting with two following objectives is performed based on algorithm 2, and the algorithm assigns a Pareto number to each feature. In step 25, we set up our secondary measure to sort the features with the same Pareto number. To do this, the crowding distance of each feature is calculated in the bi-objective space and stored in vector  $d$ .

In the last step of Algorithm 2, first, we normalize the crowding distance of features to a value in the interval  $[0, 1]$  and then set a score to each feature based on the following equation:

$$R = \text{ParetoNum} + 1/(1 + d) \quad (11)$$

Now we can sort the features based on their value in  $R$  in ascending order and store the results in a  $w$  vector that the user can select a desired number of features.

In Fig. 5, we used a sample data set with 20 features and 3 instances to show our proposed algorithm's steps with an example.

### 5. Experimental studies

To measure the performance of our proposed, we compared our method with four base FS methods which we used in the ensemble process: Fisher-Score (Duda et al., 2001), MIC (Duda et al., 2001), LLCFS (Zeng & Cheung, 2011) and CFS (Michalak & Kwasnicka, 2010). We have also compared our method to seven ensemble FS method: E-Borda (Drotár et al., 2019), E-WBorda (Drotár et al., 2019), E-Plu (Drotár et al., 2019), E-Max (Drotár et al., 2019), E-Min (Drotár et al., 2019), E-Mean (Drotár et al., 2019) and E-Median (Drotár et al., 2019).

#### 5.1. Datasets

We used 7 real-world datasets obtained to measure the performance of the proposed method with comparing methods. Table 1 contains a description of the following datasets.

#### 5.2. Performance evaluation criteria

To measure the performance of the proposed PEFS and competing methods, we used accuracy (Bolón-Canedo & Alonso-Betanzos, 2018) and F-score (Das et al., 2017) metrics. First, consider the following concepts:

- **True positive (TP):** percentage of positive examples correctly classified as so.
- **False-positive (FP):** percentage of negative examples incorrectly classified as positive.
- **True negative (TN):** percentage of negative examples correctly classified as so.
- **False-negative (FN):** percentage of positive examples incorrectly classified as negative.

The accuracy and F-score are defined as follows:

$$\text{Accuracy} = \frac{TP + TN}{TP + TN + FP + FN} \quad (12)$$

$$F - \text{score} = \frac{2 \times TP}{2 + TP + FP + FN} \quad (13)$$

#### 5.3. Results

For all comparing methods, the value of each parameter is set based on the recommendations by that corresponding paper. K-nearest neighbors (K-NN) classifier is used to compare the classification performance of the comparing algorithms and considered the number of

neighbors equal to 5. For each test, randomly, 60% of the samples are chosen as training data and the remaining 40% considered as test data. The reported results are averaged results achieved by 30 separate runs for each method; for testing each method, as the user determines the number of selected features, we change the size of features subset from 10 to 100, which results in 100 different runs on each dataset. In our method, the number of features is determined by the user. All the computations have been done on windows 8.1-64bit machine with Intel® Core (TM) i5-M460 and 4 GB Ram, using MATLAB® 9.4.0.813654 (R2018a).

### 5.3.1. Comparison between the proposed method and based FS methods

Figs. 6 and 7 show the classification performance for accuracy and F-score criteria comparing based FS methods. In these figures, the horizontal axis shows the number of selected features, and the vertical axis represents comparison criteria.

The obtained results of PEFS and all comparing methods are also compared statistically. For this purpose, The Friedman test (Hastie, Tibshirani, Friedman, & Franklin, 2017) is applied to the obtained results. The desired significance level for the post-hoc test is set to 0.05. If the result obtained by the Friedman test is less than the significance level, we perform a test for pairwise comparison of variables according to Conover (Coakley & Conover, 2000). The Friedman test results on PEFS against the other methods on 7 datasets are shown in the last row of the Table A1 of Appendix A. The values show the obtained p-values of each method, comparing the PEFS method by the Friedman test. Also, the (+) sign indicates that our approach statistically wins the comparing method, and respectively signs (=) and (-) show the tie and loss. The number of overall win/tie/loss of our method against others is shown in Table A2 of Appendix A.

### 5.3.2. Comparison between the proposed method and ensemble FS methods

The classification performance for accuracy and F-score criteria comparing ensemble FS methods is presented in Appendix A. The obtained results of PEFS and all comparing methods are also compared statistically. For this purpose, The Friedman test (Hastie et al., 2017) is applied to the obtained results. The desired significance level for the post-hoc test is set to 0.05. If the result obtained by the Friedman test is less than the significance level, we perform a test for pairwise comparison of variables according to Conover (Coakley & Conover, 2000). The Friedman test results on PEFS against the other methods on 7 datasets are shown in the last row of Tables A3–A16 in Appendix A. In the last row of tables, the p-values of each method comparing the PEFS method obtained by the Friedman test are recorded. Also, (+) in the last row of each table indicates that our method statistically wins the comparing method and respectively signs (=) and (-) show the tie and loss. Finally, we showed the number of overall win/tie/loss of our method against others in Tables A15 and A16 of Appendix A represents the run-time of algorithms.

## 5.4. Discussions

In this method, we treated the problem of ensemble feature selection

as a bi-objective optimization approach. We considered the FS methods as our decision matrix and tried to construct a ranking system for features by the concept of Pareto dominance. We tried to use a combination of redundancy-based and relevancy-based FS methods to achieve the most relevant and less redundant feature subset. To do this, we used the CFS and LLCFS method as our redundancy-based method and the other FS methods as our relevancy-based metrics. So our decision matrix is a combination of redundancy and relevancy. We used two objectives for this optimization. These two objectives are the average and minimum values assigned by FS methods to features.

We classified the results of the proposed method compared to other methods into two groups. The first group is the results of comparing the proposed method against based FS methods. The results of this category are presented by Figs. 6 and 7. The second group is the results of comparing the proposed method against ensemble FS methods. These results are also presented in Appendix A. Also, the overall win/tie/loss are presented in Table A15, and the averaged run-time of algorithms is recorded in Table A16. The results show that our method is superior to other methods in all evaluation criteria. According to these results, the PEFS method is swift, and according to the values in Table A16, the proposed method is so much faster than all ensemble methods.

## 6. Conclusion and future work

In this work, we proposed a new feature selection algorithm for ensemble learning called, the PEFS which maps the features selection process to a Pareto-based procedure. This method is based on a filter-based strategy and used a heterogeneous approach for ensemble learning. In this method, after we obtained the ranks of features based on multiple FS methods and construct a decision matrix, we deliver this data as our decision-making data to a non-dominated sorting method. The results of different datasets show the optimality and efficiency of the proposed method. We intend to use other approaches for ensemble FS, especially graph-based approaches. We also try to extend our work to different types of feature selection, like online feature selection.

### CRedit authorship contribution statement

**Amin Hashemi:** Conceptualization, Methodology, Software. **Mohammad Bagher Dowlatshahi:** Supervision, Data curation, Investigation, Writing - review & editing. **Hossein Nezamabadi-pour:** Supervision, Validation, Writing - review & editing.

### Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

## Appendix A

**Table A1**

The obtained p-values by the Friedman test in terms of accuracy and F-Score against based FS methods.

| Dataset | PEFSVS CFS |           | PEFSVS LLCFS |           | PEFSVS Fisher-Score |           | PEFSVS MIC |           |
|---------|------------|-----------|--------------|-----------|---------------------|-----------|------------|-----------|
|         | Accuracy   | F-score   | Accuracy     | F-score   | Accuracy            | F-score   | Accuracy   | F-score   |
| Colon   | 0.0005(+)  | 0.0005(+) | 0.0005(+)    | 0.0005(+) | 0.0005(+)           | 0.0005(+) | 0.0004(+)  | 0.0005(+) |
| Glioma  | 0.0005(+)  | 0.0005(+) | 0.0005(+)    | 0.0005(+) | 0.0005(+)           | 0.0005(+) | 0.0005(+)  | 0.0005(+) |
| DLBCL   | 0.0005(+)  | 0.0005(+) | 0.0005(+)    | 0.0005(+) | 0.0018(+)           | 0.0005(+) | 0.0005(+)  | 0.0005(+) |
| ORL     | 0.0005(+)  | 0.0018(+) | 0.0005(+)    | 0.0005(+) | 0.0005(+)           | 0.0005(+) | 0.008(+)   | 0.0018(+) |
| Lung    | 0.0005(+)  | 0.0005(+) | 0.0005(+)    | 0.0005(+) | 0.0005(+)           | 0.0005(+) | 0.0005(+)  | 0.0018(+) |
| Semeion | 0.0005(+)  | 0.0005(+) | 0.0056(+)    | 0.0056(+) | 0.0153(+)           | 0.0153(+) | 0.0056(+)  | 0.0056(+) |
| Shipp   | 0.0005(+)  | 0.0005(+) | 0.0004(+)    | 0.0005(+) | 0.0018(+)           | 0.0005(+) | 0.008(+)   | 0.0056(+) |

**Table A2**

The win/tie/loss results of PEFS against the based FS methods based on Friedman test

| Metric   | PEFS method against |              |        |        |
|----------|---------------------|--------------|--------|--------|
|          | CFS                 | Fisher-Score | LLCFS  | MIC    |
| Accuracy | 7/0/0               | 7/0/0        | 7/0/0  | 7/0/0  |
| F-Score  | 7/0/0               | 7/0/0        | 7/0/0  | 7/0/0  |
| Total    | 14/0/0              | 14/0/0       | 14/0/0 | 14/0/0 |

**Table A3**

Accuracy for Glioma dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.6200</b> | 0.5150            | 0.5050            | 0.4550            | 0.5050            | 0.5050            | 0.5050            | 0.5050            |
| 20            | <b>0.6650</b> | 0.5850            | 0.5200            | 0.4800            | 0.6250            | 0.6250            | 0.6250            | 0.6250            |
| 30            | <b>0.6950</b> | 0.5750            | 0.5450            | 0.4900            | 0.6150            | 0.6150            | 0.6150            | 0.6150            |
| 40            | <b>0.7050</b> | 0.5800            | 0.5500            | 0.5300            | 0.6600            | 0.6600            | 0.6600            | 0.6600            |
| 50            | <b>0.6900</b> | 0.6250            | 0.5650            | 0.5400            | 0.6550            | 0.6550            | 0.6550            | 0.6550            |
| 60            | <b>0.7050</b> | 0.6300            | 0.5500            | 0.5450            | 0.6800            | 0.6800            | 0.6800            | 0.6800            |
| 70            | <b>0.7100</b> | 0.6350            | 0.5850            | 0.5700            | 0.6750            | 0.6750            | 0.6750            | 0.6750            |
| 80            | <b>0.7150</b> | 0.6550            | 0.5950            | 0.5950            | 0.6550            | 0.6550            | 0.6550            | 0.6550            |
| 90            | <b>0.7400</b> | 0.6250            | 0.5850            | 0.5700            | 0.6700            | 0.6700            | 0.6700            | 0.6700            |
| 100           | <b>0.7100</b> | 0.6350            | 0.6200            | 0.6150            | 0.6650            | 0.6650            | 0.6650            | 0.6650            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0018 (+)</b> | <b>0.0018 (+)</b> | <b>0.0018 (+)</b> | <b>0.0018 (+)</b> |

**Table A4**

Accuracy for Colon dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.7833</b> | 0.6375            | 0.8042            | 0.7750            | 0.7708            | 0.7708            | 0.7708            | 0.7708            |
| 20            | <b>0.8208</b> | 0.6292            | 0.7708            | 0.7792            | 0.8083            | 0.8083            | 0.8083            | 0.8083            |
| 30            | 0.8167        | 0.6458            | 0.7500            | 0.8125            | <b>0.8333</b>     | <b>0.8333</b>     | <b>0.8333</b>     | <b>0.8333</b>     |
| 40            | <b>0.8375</b> | 0.6458            | 0.7667            | 0.8083            | 0.8333            | 0.8333            | 0.8333            | 0.8333            |
| 50            | <b>0.8417</b> | 0.6417            | 0.7500            | 0.8282            | 0.8250            | 0.8250            | 0.8250            | 0.8250            |
| 60            | <b>0.8417</b> | 0.6542            | 0.7542            | 0.8208            | 0.8250            | 0.8250            | 0.8250            | 0.8250            |
| 70            | <b>0.8458</b> | 0.6750            | 0.7583            | 0.8333            | 0.8125            | 0.8125            | 0.8125            | 0.8125            |
| 80            | <b>0.8500</b> | 0.6667            | 0.7750            | 0.8208            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| 90            | <b>0.8500</b> | 0.6833            | 0.7792            | 0.8125            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| 100           | <b>0.8500</b> | 0.7042            | 0.7958            | 0.8333            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0016 (+)</b> | <b>0.0005 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> |

**Table A5**

Accuracy for Lung dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.7833</b> | 0.6375            | 0.8042            | 0.7750            | 0.7708            | 0.7708            | 0.7708            | 0.7708            |
| 20            | <b>0.8208</b> | 0.6292            | 0.7708            | 0.7792            | 0.8083            | 0.8083            | 0.8083            | 0.8083            |
| 30            | 0.8167        | 0.6458            | 0.7500            | 0.8125            | <b>0.8333</b>     | <b>0.8333</b>     | <b>0.8333</b>     | <b>0.8333</b>     |
| 40            | <b>0.8375</b> | 0.6458            | 0.7667            | 0.8083            | 0.8333            | 0.8333            | 0.8333            | 0.8333            |
| 50            | <b>0.8417</b> | 0.6417            | 0.7500            | 0.8282            | 0.8250            | 0.8250            | 0.8250            | 0.8250            |
| 60            | <b>0.8417</b> | 0.6542            | 0.7542            | 0.8208            | 0.8250            | 0.8250            | 0.8250            | 0.8250            |
| 70            | <b>0.8458</b> | 0.6750            | 0.7583            | 0.8333            | 0.8125            | 0.8125            | 0.8125            | 0.8125            |
| 80            | <b>0.8500</b> | 0.6667            | 0.7750            | 0.8208            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| 90            | <b>0.8500</b> | 0.6833            | 0.7792            | 0.8125            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| 100           | <b>0.8500</b> | 0.7042            | 0.7958            | 0.8333            | 0.8167            | 0.8167            | 0.8167            | 0.8167            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0016 (+)</b> | <b>0.0005 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> | <b>0.0133 (+)</b> |

**Table A6**

Accuracy for ORL dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8630</b> | 0.8148            | 0.8407            | 0.8185            | 0.8481            | 0.8481            | 0.8481            | 0.8481            |
| 20            | <b>0.8889</b> | 0.8420            | 0.8605            | 0.8704            | 0.8741            | 0.8741            | 0.8741            | 0.8741            |
| 30            | <b>0.8988</b> | 0.8667            | 0.8580            | 0.8778            | 0.8926            | 0.8926            | 0.8926            | 0.8926            |
| 40            | <b>0.9111</b> | 0.8753            | 0.8790            | 0.8815            | 0.9074            | 0.9074            | 0.9074            | 0.9074            |
| 50            | <b>0.9210</b> | 0.8926            | 0.8778            | 0.8815            | 0.9099            | 0.9099            | 0.9099            | 0.9099            |
| 60            | <b>0.9235</b> | 0.8914            | 0.8926            | 0.8988            | 0.9136            | 0.9136            | 0.9136            | 0.9136            |
| 70            | <b>0.9272</b> | 0.8963            | 0.9086            | 0.9000            | 0.9185            | 0.9185            | 0.9185            | 0.9185            |
| 80            | <b>0.9321</b> | 0.8988            | 0.9160            | 0.9062            | 0.9259            | 0.9259            | 0.9259            | 0.9259            |
| 90            | <b>0.9321</b> | 0.9074            | 0.9160            | 0.9074            | 0.9235            | 0.9235            | 0.9235            | 0.9235            |
| 100           | <b>0.9346</b> | 0.9074            | 0.9247            | 0.9074            | 0.9235            | 0.9235            | 0.9235            | 0.9235            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0016 (+)</b> | <b>0.0051 (+)</b> | <b>0.0051 (+)</b> | <b>0.0051 (+)</b> | <b>0.0051 (+)</b> |

**Table A7**

Accuracy for DLBCL dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8883</b> | 0.7433            | 0.8133            | 0.8433            | 0.7967            | 0.7967            | 0.7967            | 0.7967            |
| 20            | <b>0.9067</b> | 0.7833            | 0.8167            | 0.8700            | 0.8200            | 0.8200            | 0.8200            | 0.8200            |
| 30            | <b>0.9033</b> | 0.7733            | 0.8367            | 0.8967            | 0.8400            | 0.8400            | 0.8400            | 0.8400            |
| 40            | 0.9200        | 0.7633            | 0.8200            | <b>0.9200</b>     | 0.8267            | 0.8267            | 0.8267            | 0.8267            |
| 50            | 0.9267        | 0.7533            | 0.8267            | <b>0.9333</b>     | 0.8267            | 0.8267            | 0.8267            | 0.8267            |
| 60            | 0.9300        | 0.7667            | 0.8267            | <b>0.9300</b>     | 0.8333            | 0.8333            | 0.8333            | 0.8333            |
| 70            | <b>0.9300</b> | 0.7900            | 0.8067            | 0.9267            | 0.8400            | 0.8400            | 0.8400            | 0.8400            |
| 80            | <b>0.9367</b> | 0.7733            | 0.8100            | 0.9067            | 0.8533            | 0.8533            | 0.8533            | 0.8533            |
| 90            | <b>0.9233</b> | 0.7833            | 0.8300            | 0.9133            | 0.8500            | 0.8500            | 0.8500            | 0.8500            |
| 100           | 0.9200        | 0.7967            | 0.8433            | <b>0.9200</b>     | 0.8433            | 0.8433            | 0.8433            | 0.8433            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.1079 (=)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A8**

Accuracy for Semeion dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8564</b> | 0.6958            | 0.7391            | 0.8378            | 0.5042            | 0.5042            | 0.5042            | 0.5042            |
| 20            | <b>0.8840</b> | 0.7783            | 0.8702            | 0.8721            | 0.5500            | 0.5500            | 0.5500            | 0.5500            |
| 30            | <b>0.9085</b> | 0.8251            | 0.9003            | 0.8954            | 0.5792            | 0.5792            | 0.5792            | 0.5792            |
| 40            | <b>0.9171</b> | 0.8440            | 0.9100            | 0.9096            | 0.5575            | 0.5575            | 0.5575            | 0.5575            |
| 50            | <b>0.9246</b> | 0.8724            | 0.9174            | 0.9220            | 0.6042            | 0.6042            | 0.6042            | 0.6042            |
| 60            | <b>0.9338</b> | 0.8793            | 0.9223            | 0.9331            | 0.6083            | 0.6083            | 0.6083            | 0.6083            |
| 70            | 0.9382        | 0.8895            | 0.9254            | <b>0.9425</b>     | 0.6083            | 0.6083            | 0.6083            | 0.6083            |
| 80            | 0.8900        | 0.8981            | 0.9287            | <b>0.9440</b>     | 0.6208            | 0.6208            | 0.6208            | 0.6208            |
| 90            | 0.8967        | 0.9064            | 0.9279            | <b>0.9469</b>     | 0.6208            | 0.6208            | 0.6208            | 0.6208            |
| 100           | 0.8933        | 0.9097            | 0.8256            | <b>0.9493</b>     | 0.6125            | 0.6125            | 0.6125            | 0.6125            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0056 (+)</b> | <b>0.1659 (=)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A9**

Accuracy for Shipp dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8533</b> | 0.7133            | 0.7167            | 0.8233            | 0.7933            | 0.7933            | 0.7933            | 0.7933            |
| 20            | 0.8800        | 0.7500            | 0.7300            | <b>0.8733</b>     | 0.8067            | 0.8067            | 0.8067            | 0.8067            |
| 30            | <b>0.8967</b> | 0.7500            | 0.7667            | 0.8533            | 0.8200            | 0.8200            | 0.8200            | 0.8200            |
| 40            | <b>0.8833</b> | 0.7667            | 0.7833            | 0.8567            | 0.8367            | 0.8367            | 0.8367            | 0.8367            |
| 50            | <b>0.8867</b> | 0.7600            | 0.7967            | 0.8633            | 0.8200            | 0.8200            | 0.8200            | 0.8200            |
| 60            | <b>0.8933</b> | 0.7833            | 0.7933            | 0.8600            | 0.8200            | 0.8200            | 0.8200            | 0.8200            |
| 70            | <b>0.8833</b> | 0.7767            | 0.7867            | 0.8767            | 0.8400            | 0.8400            | 0.8400            | 0.8400            |
| 80            | <b>0.8900</b> | 0.7867            | 0.7900            | 0.8883            | 0.8433            | 0.8433            | 0.8433            | 0.8433            |
| 90            | <b>0.8967</b> | 0.7700            | 0.7633            | 0.8700            | 0.8367            | 0.8367            | 0.8367            | 0.8367            |
| 100           | <b>0.8933</b> | 0.7733            | 0.7533            | 0.8800            | 0.8500            | 0.8500            | 0.8500            | 0.8500            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0029 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A10**

F-Score for Glioma dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.5961</b> | 0.5381            | 0.5288            | 0.4515            | 0.5371            | 0.5371            | 0.5371            | 0.5371            |
| 20            | <b>0.6720</b> | 0.6089            | 0.5303            | 0.4797            | 0.6213            | 0.6213            | 0.6213            | 0.6213            |
| 30            | <b>0.6993</b> | 0.5969            | 0.5459            | 0.6338            | 0.6459            | 0.6459            | 0.6459            | 0.6459            |
| 40            | <b>0.7011</b> | 0.5944            | 0.5493            | 0.5571            | 0.6969            | 0.6969            | 0.6969            | 0.6969            |
| 50            | <b>0.6970</b> | 0.6236            | 0.5443            | 0.5630            | 0.6925            | 0.6925            | 0.6925            | 0.6925            |
| 60            | <b>0.7081</b> | 0.6327            | 0.5695            | 0.5741            | 0.6329            | 0.6329            | 0.6329            | 0.6329            |
| 70            | 0.7108        | 0.6440            | 0.5829            | 0.5691            | <b>0.7305</b>     | <b>0.7305</b>     | <b>0.7305</b>     | <b>0.7305</b>     |
| 80            | <b>0.7208</b> | 0.6668            | 0.5959            | 0.6290            | 0.6952            | 0.6952            | 0.6952            | 0.6952            |
| 90            | <b>0.7454</b> | 0.6354            | 0.5938            | 0.5690            | 0.6683            | 0.6683            | 0.6683            | 0.6683            |
| 100           | 0.7121        | 0.6411            | 0.6345            | <b>0.7290</b>     | 0.6897            | 0.6897            | 0.6897            | 0.6897            |
| Friedman Test |               | <b>0.0018 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0833 (=)</b> | <b>0.0833 (=)</b> | <b>0.0833 (=)</b> | <b>0.0833 (=)</b> |

**Table A11**

F-score for colon dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.7713</b> | 0.5939            | 0.8002            | 0.8013            | 0.7560            | 0.7560            | 0.7560            | 0.7560            |
| 20            | <b>0.8109</b> | 0.6131            | 0.7582            | 0.7090            | 0.7928            | 0.7928            | 0.7928            | 0.7928            |
| 30            | 0.8024        | 0.6165            | 0.7367            | 0.8164            | <b>0.8223</b>     | <b>0.8223</b>     | <b>0.8223</b>     | <b>0.8223</b>     |
| 40            | <b>0.8226</b> | 0.6117            | 0.7573            | 0.8167            | 0.8207            | 0.8207            | 0.8207            | 0.8207            |
| 50            | <b>0.8275</b> | 0.6068            | 0.7390            | 0.8093            | 0.8109            | 0.8109            | 0.8109            | 0.8109            |
| 60            | <b>0.8251</b> | 0.6182            | 0.7388            | 0.7990            | 0.8093            | 0.8093            | 0.8093            | 0.8093            |
| 70            | <b>0.8323</b> | 0.6607            | 0.7480            | 0.8031            | 0.7969            | 0.7969            | 0.7969            | 0.7969            |
| 80            | <b>0.8346</b> | 0.6403            | 0.7651            | 0.7982            | 0.8016            | 0.8016            | 0.8016            | 0.8016            |
| 90            | <b>0.8359</b> | 0.6543            | 0.7693            | 0.8010            | 0.8028            | 0.8028            | 0.8028            | 0.8028            |
| 100           | <b>0.8365</b> | 0.6895            | 0.7868            | 0.7895            | 0.8010            | 0.8010            | 0.8010            | 0.8010            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0018 (+)</b> | <b>0.0377 (+)</b> | <b>0.0153 (+)</b> | <b>0.0153 (+)</b> | <b>0.0153 (+)</b> | <b>0.0153 (+)</b> |

**Table A12**

F-score for lung dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.7451</b> | 0.5669            | 0.7035            | 0.7381            | 0.6753            | 0.6753            | 0.6753            | 0.6753            |
| 20            | 0.7738        | 0.6277            | <b>0.7863</b>     | 0.7638            | 0.7752            | 0.7752            | 0.7752            | 0.7752            |
| 30            | 0.7937        | 0.6935            | 0.7593            | 0.7801            | <b>0.8047</b>     | <b>0.8047</b>     | <b>0.8047</b>     | <b>0.8047</b>     |
| 40            | 0.8272        | 0.6987            | 0.6973            | 0.7785            | <b>0.8391</b>     | <b>0.8391</b>     | <b>0.8391</b>     | <b>0.8391</b>     |
| 50            | 0.8498        | 0.7286            | 0.8070            | 0.7809            | <b>0.8636</b>     | <b>0.8636</b>     | <b>0.8636</b>     | <b>0.8636</b>     |
| 60            | <b>0.8615</b> | 0.7135            | 0.8435            | 0.7971            | 0.7598            | 0.7598            | 0.7598            | 0.7598            |
| 70            | <b>0.8683</b> | 0.7334            | 0.8617            | 0.7984            | 0.8660            | 0.8660            | 0.8660            | 0.8660            |
| 80            | 0.8826        | 0.7381            | 0.8634            | 0.8237            | <b>0.8745</b>     | <b>0.8745</b>     | <b>0.8745</b>     | <b>0.8745</b>     |
| 90            | <b>0.8792</b> | 0.7445            | 0.8620            | 0.8031            | 0.8764            | 0.8764            | 0.8764            | 0.8764            |
| 100           | <b>0.8832</b> | 0.7459            | 0.8731            | 0.7994            | 0.8751            | 0.8751            | 0.8751            | 0.8751            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0153 (+)</b> | <b>0.0018 (+)</b> | <b>0.4884 (+)</b> | <b>0.4884 (+)</b> | <b>0.4884 (+)</b> | <b>0.4884 (+)</b> |

**Table A13**

F-Score for ORL dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.5185</b> | 0.4438            | 0.4701            | 0.4053            | 0.4208            | 0.4208            | 0.4208            | 0.4208            |
| 20            | <b>0.5743</b> | 0.5588            | 0.5724            | 0.4098            | 0.4930            | 0.4930            | 0.4930            | 0.4930            |
| 30            | <b>0.6410</b> | 0.5960            | 0.6153            | 0.4102            | 0.5158            | 0.5158            | 0.5158            | 0.5158            |
| 40            | <b>0.6621</b> | 0.6166            | 0.6497            | 0.5317            | 0.5357            | 0.5357            | 0.5357            | 0.5357            |
| 50            | <b>0.6904</b> | 0.6337            | 0.6699            | 0.5790            | 0.5591            | 0.5591            | 0.5591            | 0.5591            |
| 60            | <b>0.6983</b> | 0.6615            | 0.6736            | 0.5799            | 0.5606            | 0.5606            | 0.5606            | 0.5606            |
| 70            | <b>0.7075</b> | 0.6719            | 0.6911            | 0.6085            | 0.5737            | 0.5737            | 0.5737            | 0.5737            |
| 80            | <b>0.7003</b> | 0.6780            | 0.6862            | 0.6207            | 0.5766            | 0.5766            | 0.5766            | 0.5766            |
| 90            | <b>0.7115</b> | 0.6885            | 0.7032            | 0.5941            | 0.5847            | 0.5847            | 0.5847            | 0.5847            |
| 100           | <b>0.7290</b> | 0.6845            | 0.7044            | 0.6213            | 0.6034            | 0.6034            | 0.6034            | 0.6034            |
| Friedman Test |               | <b>0.0018 (+)</b> | <b>0.0056 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A14**

F-Score DLBCL dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8442</b> | 0.6651            | 0.3160            | 0.6980            | 0.6633            | 0.6633            | 0.6633            | 0.6633            |
| 20            | <b>0.8780</b> | 0.7104            | 0.3160            | 0.7369            | 0.7155            | 0.7155            | 0.7155            | 0.7155            |
| 30            | <b>0.8757</b> | 0.7155            | 0.3160            | 0.7730            | 0.7381            | 0.7381            | 0.7381            | 0.7381            |
| 40            | <b>0.8982</b> | 0.7151            | 0.3160            | 0.7970            | 0.7545            | 0.7545            | 0.7545            | 0.7545            |
| 50            | <b>0.9104</b> | 0.7465            | 0.3160            | 0.8005            | 0.7520            | 0.7520            | 0.7520            | 0.7520            |
| 60            | <b>0.9122</b> | 0.7385            | 0.3503            | 0.8220            | 0.7559            | 0.7559            | 0.7559            | 0.7559            |
| 70            | <b>0.9124</b> | 0.7419            | 0.3980            | 0.8010            | 0.7651            | 0.7651            | 0.7651            | 0.7651            |
| 80            | <b>0.9179</b> | 0.7351            | 0.5636            | 0.7901            | 0.7629            | 0.7629            | 0.7629            | 0.7629            |
| 90            | <b>0.9094</b> | 0.7506            | 0.6857            | 0.8021            | 0.7622            | 0.7622            | 0.7622            | 0.7628            |
| 100           | <b>0.9074</b> | 0.7465            | 0.7199            | 0.8012            | 0.7641            | 0.7641            | 0.7641            | 0.7641            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0056 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |



**Table A15**

F-Score for Semeion dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8568</b> | 0.5719            | 0.5568            | 0.8320            | 0.5903            | 0.5903            | 0.5903            | 0.5903            |
| 20            | <b>0.8841</b> | 0.4971            | 0.5109            | 0.8450            | 0.5567            | 0.5567            | 0.5567            | 0.5567            |
| 30            | <b>0.9086</b> | 0.5098            | 0.4226            | 0.9028            | 0.6049            | 0.6049            | 0.6049            | 0.6049            |
| 40            | <b>0.9174</b> | 0.5423            | 0.4437            | 0.9102            | 0.6087            | 0.6087            | 0.6087            | 0.6087            |
| 50            | <b>0.9250</b> | 0.5481            | 0.4160            | 0.9236            | 0.6143            | 0.6143            | 0.6143            | 0.6143            |
| 60            | <b>0.9340</b> | 0.5231            | 0.4150            | 0.9330            | 0.6022            | 0.6022            | 0.6022            | 0.6022            |
| 70            | 0.9385        | 0.5086            | 0.4403            | <b>0.9480</b>     | 0.5550            | 0.5550            | 0.5550            | 0.5550            |
| 80            | 0.9445        | 0.5584            | 0.4345            | <b>0.9550</b>     | 0.5777            | 0.5777            | 0.5777            | 0.5777            |
| 90            | 0.9483        | 0.5296            | 0.4241            | <b>0.9490</b>     | 0.5840            | 0.5840            | 0.5840            | 0.5840            |
| 100           | 0.9502        | 0.5245            | 0.4571            | <b>0.9530</b>     | 0.5317            | 0.5317            | 0.5317            | 0.5317            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0056 (+)</b> | <b>0.1659 (=)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A16**

F-Score for Shipp dataset.

| M/feat        | PEFS          | E-Bords           | E-WBords          | E-Plu             | E-Max             | E-Min             | E-Median          | E-Mean            |
|---------------|---------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|-------------------|
| 10            | <b>0.8356</b> | 0.5854            | 0.6056            | 0.7905            | 0.7307            | 0.7307            | 0.7307            | 0.7307            |
| 20            | <b>0.8623</b> | 0.6289            | 0.6007            | 0.8451            | 0.7482            | 0.7482            | 0.7482            | 0.7482            |
| 30            | <b>0.8852</b> | 0.6531            | 0.6646            | 0.8401            | 0.7660            | 0.7660            | 0.7660            | 0.7660            |
| 40            | <b>0.8671</b> | 0.6829            | 0.6871            | 0.8398            | 0.7830            | 0.7830            | 0.7830            | 0.7830            |
| 50            | <b>0.8696</b> | 0.6849            | 0.7269            | 0.8325            | 0.7562            | 0.7562            | 0.7562            | 0.7562            |
| 60            | <b>0.8827</b> | 0.7341            | 0.7246            | 0.8520            | 0.7400            | 0.7400            | 0.7400            | 0.7400            |
| 70            | <b>0.8741</b> | 0.7298            | 0.6988            | 0.8520            | 0.7734            | 0.7734            | 0.7734            | 0.7734            |
| 80            | <b>0.8804</b> | 0.9408            | 0.7062            | 0.8520            | 0.7781            | 0.7781            | 0.7781            | 0.7781            |
| 90            | <b>0.8856</b> | 0.7135            | 0.6709            | 0.8730            | 0.7701            | 0.7701            | 0.7701            | 0.7701            |
| 100           | <b>0.8801</b> | 0.7089            | 0.6675            | 0.8603            | 0.7985            | 0.7985            | 0.7985            | 0.7985            |
| Friedman Test |               | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0029 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> | <b>0.0005 (+)</b> |

**Table A17**

The win/tie/loss results of PEFS against the ensemble FS methods based on Friedman test.

| Metric   | PEFS method against |          |        |        |        |          |        |
|----------|---------------------|----------|--------|--------|--------|----------|--------|
|          | E-Borda             | E-WBorda | E-Plu  | E-Min  | E-Max  | E-Median | E-Mean |
| Accuracy | 7/0/0               | 7/0/0    | 5/2/0  | 7/0/0  | 7/0/0  | 7/0/0    | 7/0/0  |
| F-Score  | 7/0/0               | 7/0/0    | 6/1/0  | 5/2/0  | 5/2/0  | 5/2/0    | 5/2/0  |
| Total    | 14/0/0              | 14/0/0   | 11/3/0 | 12/2/0 | 12/2/0 | 12/2/0   | 12/2/0 |

**Table A18**

The average runtime of PEFS against the ensemble FS methods.

| Datasets | Methods       |         |          |       |       |       |          |        |
|----------|---------------|---------|----------|-------|-------|-------|----------|--------|
|          | PEFS          | E-Borda | E-WBorda | E-Plu | E-Min | E-Max | E-Median | E-Mean |
| Colon    | <b>1.84</b>   | 1220    | 1110     | 1114  | 1105  | 1101  | 1108     | 1106   |
| Glioma   | <b>36.74</b>  | 1670    | 1791     | 1688  | 1680  | 1671  | 1674     | 1676   |
| DLBCL    | <b>9.30</b>   | 1670    | 1791     | 1688  | 1680  | 1671  | 1674     | 1676   |
| ORL      | <b>200.84</b> | 2008    | 2013     | 2010  | 2008  | 2007  | 2009     | 2010   |
| Lung     | <b>87.78</b>  | 3059    | 3063     | 3057  | 3053  | 3050  | 3059     | 3060   |
| Semeion  | <b>57.78</b>  | 3763    | 3770     | 3760  | 3759  | 3757  | 3761     | 3762   |
| Shipp    | <b>76.55</b>  | 3250    | 3441     | 2775  | 2774  | 2770  | 2698     | 2778   |

## References

- Ansari, G., Ahmad, T., & Doja, M. N. (2019). Ensemble of feature ranking methods using hesitant fuzzy sets for sentiment classification. *International Journal of Machine Learning and Computing*, 9(5), 599–608.
- Bache, K., & Lichman, M. (2013). *Repository, UCI machine learning*. Irvine, CA: University of California.
- Bayati, H., Dowlatshahi, M. B., & Paniri, M. (2020). MLPSO: A filter multi-label feature selection based on particle swarm optimization. In *2020 25th International Computer Conference, Computer Society of Iran (CSICC)* (pp. 1–6). <https://doi.org/10.1109/CSICC49403.2020.9050087>
- Ben Brahim, A., & Limam, M. (2018). Ensemble feature selection for high dimensional data: A new method and a comparative study. *Advances in Data Analysis and Classification*, 12(4), 937–952. <https://doi.org/10.1007/s11634-017-0285-y>
- Bolón-Canedo, V., & Alonso-Betanzos, A. (2018). Evaluation of ensembles for feature selection. In *Intelligent Systems Reference Library* (Vol. 147, pp. 97–113). [https://doi.org/10.1007/978-3-319-90080-3\\_6](https://doi.org/10.1007/978-3-319-90080-3_6).
- Bolón-Canedo, V., & Alonso-Betanzos, A. (2019). Ensembles for feature selection: A review and future trends. *Information Fusion*, 52, 1–12. <https://doi.org/10.1016/j.inffus.2018.11.008>
- Bozók, S., & Fülöp, J. (2018). Efficient weight vectors from pairwise comparison matrices. *European Journal of Operational Research*, 264(2), 419–427. <https://doi.org/10.1016/j.ejor.2017.06.033>
- Cai, J., Luo, J., Wang, S., & Yang, S. (2018). Feature selection in machine learning: A new perspective. *Neurocomputing*, 300, 70–79. <https://doi.org/10.1016/j.neucom.2017.11.077>
- Coakley, C. W., & Conover, W. J. (2000). Practical Nonparametric Statistics. *Journal of the American Statistical Association*, 95(449), 332. <https://doi.org/10.2307/2669565>
- Das, A. K., Das, S., & Ghosh, A. (2017). Ensemble feature selection using bi-objective genetic algorithm. *Knowledge-Based Systems*, 123, 116–127. <https://doi.org/10.1016/j.knsys.2017.02.013>
- Dowlatshahi, M. B., Derhami, V., & Nezamabadi-Pour, H. (2020). Fuzzy particle swarm optimization with nearest-better neighborhood for multimodal optimization. *Iranian Journal of Fuzzy Systems*, 17(4), 7–24.

- Dowlatshahi, M. B., & Derhami, V. (2017). Winner determination in combinatorial auctions using hybrid ant colony optimization and multi-neighborhood local search. *Journal of AI and Data Mining*, 5(2), 169–181.
- Dowlatshahi, M. B., Derhami, V., & Nezamabadi-pour, H. (2018). A novel three-stage filter-wrapper framework for miRNA subset selection in cancer classification. *Informatics*, 5(1), 13. <https://doi.org/10.3390/informatics5010013>
- Dowlatshahi, M., Derhami, V., & Nezamabadi-pour, H. (2017). Ensemble of filter-based rankers to guide an epsilon-greedy swarm optimizer for high-dimensional feature subset selection. *Information (Switzerland)*, 8(4), 152. <https://doi.org/10.3390/info8040152>
- Dowlatshahi, M. B., & Nezamabadi-Pour, H. (2014). GGSA: A grouping gravitational search algorithm for data clustering. *Engineering Applications of Artificial Intelligence*, 36, 114–121. <https://doi.org/10.1016/j.engappai.2014.07.016>
- Dowlatshahi, M. B., Nezamabadi-Pour, H., & Mashinchi, M. (2014). A discrete gravitational search algorithm for solving combinatorial optimization problems. *Information Sciences*, 258, 94–107. <https://doi.org/10.1016/j.ins.2013.09.034>
- Dowlatshahi, M. B., & Rezaeian, M. (2016). Training spiking neurons with gravitational search algorithm for data classification. In *1st Conference on Swarm Intelligence and Evolutionary Computation*. <https://doi.org/10.1109/CSIEC.2016.7482125>
- Drotár, P., Gazda, M., & Vokorokos, L. (2019). Ensemble feature selection using election methods and ranker clustering. *Information Sciences*, 480, 365–380. <https://doi.org/10.1016/j.ins.2018.12.033>
- Duda, R. O., Hart, P. E., & Stork, D. G. (2001). Pattern classification. New York: John Wiley, Section, 10, 1.
- Duleba, S., & Moslem, S. (2019). Examining Pareto optimality in analytic hierarchy process on real Data: An application in public transport service development. *Expert Systems with Applications*, 116, 21–30. <https://doi.org/10.1016/j.eswa.2018.08.049>
- Ebrahimpour, M. K., & Eftekhari, M. (2017). Ensemble of feature selection methods: A hesitant fuzzy sets approach. *Applied Soft Computing Journal*, 50, 300–312. <https://doi.org/10.1016/j.asoc.2016.11.021>
- Ehrgott, M. (2000). Approximation algorithms for combinatorial multicriteria optimization problems. *International Transactions in Operational Research*, 7(1), 5–31. <https://doi.org/10.1111/itor.2000.7.issue-110.1111/j.1475-3995.2000.tb00182.x>
- Gustavsson, P., & Syberfeldt, A. (2018). A new algorithm using the non-dominated tree to improve non-dominated sorting. *Evolutionary Computation*, 26(1), 89–116. [https://doi.org/10.1162/evco\\_a\\_00204](https://doi.org/10.1162/evco_a_00204)
- Hancer, E., Xue, B., & Zhang, M. (2018). Differential evolution for filter feature selection based on information theory and feature ranking. *Knowledge-Based Systems*, 140, 103–119. <https://doi.org/10.1016/j.knsys.2017.10.028>
- Hashemi, A., Dowlatshahi, M. B., & Nezamabadi-pour, H. (2020a). MGFS: A multi-label graph-based feature selection algorithm via PageRank centrality. *Expert Systems with Applications*, 142, 113024. <https://doi.org/10.1016/j.eswa.2019.113024>
- Hashemi, A., & Dowlatshahi, M. B. (2020). MLCR: A fast multi-label feature selection method based on K-means and L2-norm. In *2020 25th International Computer Conference, Computer Society of Iran (CSICC)* (pp. 1–7). <https://doi.org/10.1109/CSICC49403.2020.9050104>
- Hashemi, A., Dowlatshahi, M. B., & Nezamabadi-pour, H. (2020b). MFS-MCDM: Multi-label feature selection using multi-criteria decision making. *Knowledge-Based Systems*, 206, 106365. <https://doi.org/10.1016/j.knsys.2020.106365>
- Hashemi, A., Dowlatshahi, M. B., & Nezamabadi-Pour, H. (2021). A bipartite matching-based feature selection for multi-label learning. *International Journal of Machine Learning and Cybernetics*, 12(2), 459–475. <https://doi.org/10.1007/s13042-020-01180-w>
- Hastie, T., Tibshirani, R., Friedman, J., & Franklin, J. (2017). The Elements of Statistical Learning: Data Mining, Inference, and Prediction. Math. Intell. <https://doi.org/10.1007/BF02985802>
- Hoque, N., Singh, M., & Bhattacharyya, D. K. (2018). EFS-MI: An ensemble feature selection method for classification. *Complex & Intelligent Systems*, 4(2), 105–118. <https://doi.org/10.1007/s40747-017-0060-x>
- Jadhav, S., He, H., & Jenkins, K. (2018). Information gain directed genetic algorithm wrapper feature selection for credit rating. *Applied Soft Computing Journal*, 69, 541–553. <https://doi.org/10.1016/j.asoc.2018.04.033>
- Li, J., Cheng, K., Wang, S., Morstatter, F., Trevino, R. P., Tang, J., & Liu, H. (2018). Feature selection: A data perspective. *ACM Computing Surveys*, 50(6), 1–45. <https://doi.org/10.1145/3136625>
- Li, M., Yang, S., & Liu, X. (2015). Bi-goal evolution for many-objective optimization problems. *Artificial Intelligence*, 228, 45–65. <https://doi.org/10.1016/j.artint.2015.06.007>
- Miao, J., & Niu, L. (2016). A survey on feature selection. *Procedia Computer Science*, 91, 919–926. <https://doi.org/10.1016/j.procs.2016.07.111>
- Michalak, K., & Kwasnicka, H. (2010). Correlation based feature selection method. *International Journal of Bio-Inspired Computation*, 2(5), 319–332. <https://doi.org/10.1504/IJBIC.2010.036158>
- Mlambo, N. W. C. (2016). A survey and comparative study of filter and wrapper feature selection techniques. *The International Journal Of Engineering And Science (IJES)*, 5(8), 57–67. <https://pdfs.semanticscholar.org/fe2c/a244dd5473270ce0a8e2a387349b9831cf47.pdf>
- Momeni, Ehsan, Dowlatshahi, Mohammad Bagher, Omidinasab, Fereydoon, Maizir, Harnedi, & Armaghani, Danial Jahed (2020). Gaussian process regression technique to estimate the pile bearing capacity. *Arabian Journal for Science and Engineering*, 45(10), 8255–8267. <https://doi.org/10.1007/s13369-020-04683-4>
- Momeni, Ehsan, Yarivand, Akbar, Dowlatshahi, Mohammad Bagher, & Armaghani, Danial Jahed (2021). An efficient optimal neural network based on gravitational search algorithm in predicting the deformation of geogrid-reinforced soil structures. *Transportation Geotechnics*, 26, 100446. <https://doi.org/10.1016/j.trgeo.2020.100446>
- Ng, W. W. Y., Tuo, Y., Zhang, J., & Kwong, S. (2020). Training error and sensitivity-based ensemble feature selection. *International Journal of Machine Learning and Cybernetics*, 11(10), 2313–2326. <https://doi.org/10.1007/s13042-020-01120-8>
- Nouri-Moghaddam, Babak, Ghazanfari, Mehdi, & Fathian, Mohammad (2021). A novel multi-objective forest optimization algorithm for wrapper feature selection. *Expert Systems with Applications*, 175, 114737. <https://doi.org/10.1016/j.eswa.2021.114737>
- Paniri, Mohsen, Dowlatshahi, Mohammad Bagher, & Nezamabadi-pour, Hossein (2020). MLACO: A multi-label feature selection algorithm based on ant colony optimization. *Knowledge-Based Systems*, 192, 105285. <https://doi.org/10.1016/j.knsys.2019.105285>
- Pereira, R. B., Plastino, A., Zadrozny, B., & Merschmann, L. H. C. (2018). Categorizing feature selection methods for multi-label classification. *Artificial Intelligence Review*, 49(1), 57–78. <https://doi.org/10.1007/s10462-016-9516-4>
- Rafsanjani, Marjan Kuchaki, & Dowlatshahi, Mohammad Bagher (2012). Using gravitational search algorithm for finding near-optimal base station location in two-tiered WSNs. *International Journal of Machine Learning and Computing*, 377–380. <https://doi.org/10.7763/IJMLC.2012.V2.148>
- Rafsanjani, M. K., Dowlatshahi, M. B., & Nezamabadi-Pour, H. (2015). Gravitational search algorithm to solve the K-of-N lifetime problem in two-tiered WSNs. *Iranian Journal of Mathematical Sciences and Informatics*, 10(1), 81–93. <https://doi.org/10.7508/ijmsi.2015.01.006>
- Raquel, C. R., & Naval, P. C. (2005). An effective use of crowding distance in multiobjective particle swarm optimization. *GECCO 2005 - Genetic and Evolutionary Computation Conference*, 257–264. <https://doi.org/10.1145/1068009.1068047>
- Reyes, Oscar, Morell, Carlos, & Ventura, Sebastián (2015). Scalable extensions of the ReliefF algorithm for weighting and selecting features on the multi-label learning context. *Neurocomputing*, 161, 168–182. <https://doi.org/10.1016/j.neucom.2015.02.045>
- Samaria, F. S., & Harter, A. C. (1994). Parameterisation of a stochastic model for human face identification. *IEEE Workshop on Applications of Computer Vision – Proceedings*, 138–142. <https://doi.org/10.1109/acv.1994.341300>
- Seijo-Pardo, B., Porto-Díaz, I., Bolón-Canedo, V., & Alonso-Betanzos, A. (2017). Ensemble feature selection: Homogeneous and heterogeneous approaches. *Knowledge-Based Systems*, 118, 124–139. <https://doi.org/10.1016/j.knsys.2016.11.017>
- Sheikhpour, R., Sarram, M. A., Gharaghani, S., & Chahooki, M. A. Z. (2017). A Survey on semi-supervised feature selection methods. *Pattern Recognition*, 64, 141–158. <https://doi.org/10.1016/j.patcog.2016.11.003>
- Shipp, Margaret A., Ross, Ken N., Tamayo, Pablo, Weng, Andrew P., Kutok, Jeffery L., Aguiar, Ricardo C. T., ... Golub, Todd R. (2002). Diffuse large B-cell lymphoma outcome prediction by gene-expression profiling and supervised machine learning. *Nature Medicine*, 8(1), 68–74. <https://doi.org/10.1038/nm0102-68>
- Talbi, E. G. (2009). Metaheuristics: From Design to Implementation. In Metaheuristics: From Design to Implementation. <https://doi.org/10.1002/9780470496916>
- Tian, Yiming, Zhang, Jie, Wang, Jie, Geng, Yanli, & Wang, Xitai (2020). Robust human activity recognition using single accelerometer via wavelet energy spectrum features and ensemble feature selection. *Systems Science and Control Engineering*, 8(1), 83–96. <https://doi.org/10.1080/21642583.2020.1723142>
- Tsai, Chih-Pong, & Sung, Ya-Ting (2020). Ensemble feature selection in high dimension, low sample size datasets: Parallel and serial combination approaches. *Knowledge-Based Systems*, 203, 106097. <https://doi.org/10.1016/j.knsys.2020.106097>
- Venkatesh, B., & Anuradha, J. (2019). A review of Feature Selection and its methods. *Cybernetics and Information Technologies*, 19(1), 3–26. <https://doi.org/10.2478/CAIT-2019-0001>
- Von Lücken, C., Barán, B., & Brizuela, C. (2014). A survey on multi-objective evolutionary algorithms for many-objective problems. *Computational Optimization and Applications*, 58(3), 707–756. <https://doi.org/10.1007/s10589-014-9644-1>
- Zhang, Xingyi, Tian, Ye, Cheng, Ran, & Jin, Yaochu (2015). An Efficient Approach to Nondominated Sorting for Evolutionary Multiobjective Optimization. *IEEE Transactions on Evolutionary Computation*, 19(2), 201–213. <https://doi.org/10.1109/TEVC.2014.2308305>
- Xu, W., Chong, A., Karaguzel, O. T., & Lam, K. P. (2016). Improving evolutionary algorithm performance for integer type multi-objective building system design optimization. *Energy and Buildings*, 127, 714–729. <https://doi.org/10.1016/j.enbuild.2016.06.043>
- Yuan, Yuan, Xu, Hua, Wang, Bo, & Yao, Xin (2016). A new dominance relation-based evolutionary algorithm for many-objective optimization. *IEEE Transactions on Evolutionary Computation*, 20(1), 16–37. <https://doi.org/10.1109/TEVC.423510.1109/TEVC.2015.2420112>
- Zeng, H., & Cheung, Y. M. (2011). Feature selection and kernel learning for local learning-based clustering. *IEEE Transactions on Pattern Analysis and Machine Intelligence*, 33(8), 1532–1547. <https://doi.org/10.1109/TPAMI.2010.215>
- Zhang, R., Nie, F., Li, X., & Wei, X. (2019). Feature selection with multi-view data: A survey. *Information Fusion*, 50, 158–167. <https://doi.org/10.1016/j.inffus.2018.11.019>
- Zhou, HongFang, Zhang, JiaWei, Zhou, YueQing, Guo, XiaoJie, & Ma, YiMing (2021). A feature selection algorithm of decision tree based on feature weight. *Expert Systems with Applications*, 164, 113842. <https://doi.org/10.1016/j.eswa.2020.113842>
- Zhu, Q. H., & Bin Yang, Y. (2018). Discriminative embedded unsupervised feature selection. *Pattern Recognition Letters*, 112, 219–225. <https://doi.org/10.1016/j.patrec.2018.07.018>